MADISYN BROOKS

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SUMMARY

Detail-oriented physics PhD candidate and astrophysics researcher with strong technical expertise in Python, machine learning, data analysis, and statistical modeling, with additional experience in Git, Jupyter, LaTeX, and scientific computing. Proficient at communicating technical concepts through peer-reviewed publications (400+ citations), presentations, and public outreach, with a strong foundation in physics and problem solving. Relevant coursework in predictive modeling, computational physics, linear algebra, and calculus.

SKILLS

Python	NumPy, Pandas, SciPy, Matplotlib, emcee, Seaborn, AstroPy,
Statistics	Bayesian Inference, MCMC, Model Comparison (BIC), Regression Models, Survival Analysis, Robustness Testing
Machine Learning	scikit-learn, PyTorch, GMMs, DBSCAN, Dimensionality Reduction
Tools	Jupyter Notebook (Anaconda), Git, LaTeX, VSCode, Windows
Working Knowledge	IDL, HTML, C, Linux, Parallel Computing

DEEP LEARNING PROJECT

Spectral Classifier - Multimodal Variational Autoencoder, Independent Project

- Designing a **multimodal VAE** in **PyTorch** with per-modality encoder/decoder networks fused into a shared latent space, built to handle heterogeneous and partially-observed inputs
- Validating a full training pipeline: data pre-processing and normalization, missing-data handling, loss testing, and latent-space evaluation using **clustering algorithms**

RESEARCH & PROFESSIONAL EXPERIENCE

Graduate Research Assistant, University of Connecticut *2023 - Present*

- Applied unsupervised **machine learning clustering (GMMs)** techniques to identify distinct sub-populations of galaxies within large JWST catalogs
- Performed **Bayesian inference** and applied **Markov chain Monte Carlo** methods to model complex multi-parameter distributions of galaxy properties
- Applied model comparisons using the **Bayesian Information Criteria (BIC)** to evaluate different physical hypotheses and **performed statistical tests (AD, KS)** to compare underlying data distributions
- Performed **bias mitigation** by designing analysis methods that correct for selection effects in observational data
- Implemented **bootstrap resampling to estimate uncertainties** on median-stacked spectra in the absence of analytic error models
- Normalized, aligned, and aggregated large ($\sim 20,000$ galaxies) multi-dimensional datasets (spectra + photometry) into analysis-ready formats and prepared publication-quality figures
- **Three first author publications**, in peer-reviewed astrophysical journals (2 published; 1 submitted)

EDUCATION

University of Connecticut, PhD in Physics *2023 - May 2028 (expected)*

University of Central Florida, Bachelors of Science in Physics 2019 - 2023
Cum Laude